

1. Questions

In a river, two points P and Q are 66 km apart from each other and there is a point X somewhere in between them. Boat A starts from point X towards point P in upstream direction and boat B starts from point X towards point Q in downstream direction. The speeds of boats A and B in still water are 14.5 km/h and 12 km/h respectively. If boat A took 3 hours to reach point P and boat B took 2 hours to reach point Q, then find the distance between points P and X.

- a. 36 km
- b. 39 km
- c. 30 km
- d. 45 km
- e. 42 km

2. Questions

A man invested ₹15000 in scheme A, which gives interest at the rate of R% compounded half yearly. He also invested ₹10000 in scheme B, which gives interest at the rate of r% compounded half yearly. Find the ratio of the compound interest received from scheme A after 1 year to that received from scheme B after 1 year.

Values of 'R' and 'r' are respectively the bigger and the smaller roots of the equation given below:

$$15x^2 + 26x + 169 = 14x^2 + 48x + 73$$

- a. 416: 103
- b. 4: 1
- c. 208: 53
- d. 832: 203
- e. 13: 4

3. Questions

In mixture P, the quantity of type A milk is 20 L more than the quantity of type B milk. The quantity of type A milk in mixture Q is half of that in mixture P and the quantity of type B milk in mixture Q is double of that in mixture P. If the costs of type A milk and type B milk are ₹60 per L and the ₹40 per L respectively and the cost mixture P is ₹55 per L, then find that which of the following statements is/are not true.

I: The difference between the quantities of type A milk and type B milk in mixture Q is 10 L.

II: The cost of mixture Q is more than ₹50 per L.

III: The ratio of the total quantity of mixture P to that of mixture Q is 5: 4.

- a. Only I and II

- b. None is true
- c. Only III
- d. Only I and III
- e. Only II

4. Questions

A shopkeeper has 'X' units of an article and he marked up each unit by 25%. He sold each of 'X' units at 10% discount on the marked price and the total profit received is equal to the cost price of 1 unit of the article. If each unit of the article were sold at ____% discount on the marked price, there would be neither profit nor loss. Find the value, which will be filled in the blank.

- a. $(X - 5)\%$
- b. $(X + 15)\%$
- c. $(X + 12)\%$
- d. $(X - 1)\%$
- e. $(X + 10)\%$

5. Questions

In a factory, machine A manufactures 25 units per hour of an article, machine B manufactures X units per hour of the article and machine C manufactures $(X + 5)$ units per hour of the article. If the ratio of the time taken by machine A to manufacture a certain number of units to the time taken by machine C to manufacture the same number of units, is 3: 5, then find that which of the following can't be determined.

I: The ratio of the efficiency of machine A to that of machine B.

II: The time taken by machine B to manufacture 300 units of the article.

III: Find the total number of units manufactured by machines A, B and C together in 2.4 hours.

- a. Only II and III
- b. Only II
- c. Only I and III
- d. Only I
- e. All can be determined

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6. Questions

Study the following data carefully and answer the questions:

Three series I, II and III are given below and each series follows different patterns:

I: 677, 388, 219, 98, P, 24

II: $(P * Q)$, 691, 466, 297, 176, 95

III: $(2P/7)$, $(Q + 10)$, $(P + Q - 3)$, $(7Q - 10)$, $(5P - 3Q/4)$

Find the next term of series III.

- a. 374
- b. 394
- c. 354
- d. 364
- e. 384

7. Questions

Find that which of the following statements is/are not true.

X: P and Q are co-prime with each other.

Y: Difference of P and Q is a prime number.

Z: Value of $(2P + 5Q)$ is a perfect square.

- a. Only X and Z
- b. None is true
- c. Only Y
- d. Only X
- e. Only Z

8. Questions

Three series I, II and III are given below and each series follows different patterns:

I: 5, 11, 21, 44, 85, 171, 341

II: 1260, 315, 90, 30, 15, 6

III: 9, 8, 65, 24, 210, 48

"If P, Q, and R are the incorrect terms of Series I, II, and III respectively, and A, B, and C are the correct terms that should replace P, Q, and R respectively, then find the ratio of $(P + Q + R)$ to $(A + B + C)$."

- a. 23: 24
- b. 113: 136
- c. 269: 272
- d. 4: 5
- e. None of these

9. Questions

In each question, read the given statement and compare the Quantity I and Quantity II on its basis. (Only quantity is to be considered)

The ratio of A's present age to B's present age is 3: 4 and C's present age is three time of the difference between the present ages of A and B. D is older than A and the difference between the present ages of A and D is 3 years. The ratio of A's age after 9 years to C's age 6 years ago is 9: 4.

Quantity I: Find the average of the present ages of A, B and C.

Quantity II: Find D's present age.

- a. Quantity I $\geq$ Quantity II
- b. Quantity I $>$ Quantity II
- c. Quantity I $<$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II or relation can't be established

10. Questions

Car A is running towards cars B and C. Car B is following car C. Cars A and B are 780 m away from each other and cars A and C are 384 m away from each other. Car A crosses car B in 15 seconds and crosses car C in 8 seconds.

Quantity I: Find the time (in seconds), in which car B will cross car C.

Quantity II: Find the speed of car A (in km/h).

- a. Quantity I $>$ Quantity II
- b. Quantity I $<$ Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II or relation can't be established

11. Questions

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Study the following data carefully and answer the questions:

Data given below is related to the number of e-bikes and e-scooters sold by two companies A and B in five months i.e; January, February, March, April and May.

Table given below shows the following data:

Month	(No. of e-bikes sold by company A) – (No. of e-bikes sold by company B)	(No. of e-bikes – No. of e-scooter) sold by company A	(No. of e-bikes – No. of e-scooter) sold by company B
January	—	6	-5
February	21	—	-9
March	3	-7	—
April	—	—	5
May	-8	-12	—

If the ratio of the number of e-bikes to the number of e-scooters sold by company A in March is 4: 5, the ratio of the number of e-scooters sold by company A to that sold by company B in January is 3: 5 and the number of e-scooters sold by company A in January is 5 less than that sold by company A in March, then find the difference between the number of e-bikes sold by companies A and B in January.

- 9
- 5
- 12
- 15
- 7

12. Questions

The ratio of the number of e-bikes sold by company A to that sold by company B in February is 17: 10 and the number of e-bikes sold by company B in February is equal to that sold by company A in May. If the number of e-scooters sold by company B in May is 10 less than the number of e-bikes sold by company B in May, then find the ratio of the number of e-scooters sold by company A to that sold by company B in May.

- Can't be determined
- 21: 20
- 7: 4
- 6: 5
- 3: 2

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13. Questions

If the number of e-bikes sold by company B in April is $16\frac{2}{3}\%$ more than the number of e-scooters sold by company B in April and also 87.5% of the number of e-bikes sold by company A in April and the number of e-scooter sold by company A in April is 20% more than that sold by

company B in April, then find the difference between the number of e-bikes and e-scooters sold by company A in April?

- a. 6
- b. 10
- c. 2
- d. 8
- e. 4

14. Questions

If the numbers of e-bikes sold by company B in January and February are 15 and 25 respectively, the average number of e-scooters sold by company B in January, February and March is 24 and the difference between the number of e-bikes and e-scooters sold by company B in March is 3, which of the following statements definitely true?

I: The number of e-bikes sold by company B in January and March is equal.

II: The total number of e-bikes and e-scooters sold by company B in March can be less than 40.

III: The average number of e-bikes sold by company B in January, February and March can be 20.

- a. Only I and II
- b. Only II
- c. Only II and III
- d. Only I and III
- e. Only I

15. Questions

If the number of e-scooters sold by company A in February is 72.5% of the number of e-bikes sold by company A in February and the difference between them is a prime number, then find that the difference between the them in February is equal to which of the following?

- a. The difference between the number of e-bikes and e-scooters sold by company A in MayThe difference between the number of e-bikes and e-scooters sold by company B in February
- b. The difference between the number of e-bikes and e-scooters sold by company B in February
- c. The difference between the e-bikes sold by companies A and B in February
- d. None of these
- e. The difference between the e-bikes sold by companies A and B in May

16. Questions

In each of the following questions, read the given statement and compare Quantity I and Quantity

II on its basis. (Only quantity is to be considered)

Quantity I: Rahul alone can finish a work in 24 days. Along with Prateek and Saurabh, he can finish the work in 6 days. If Saurabh is 20% more efficient than Prateek, then find the time taken (days) by Saurabh alone to finish the work.

Quantity II: The average age of Manish and Mohan is 32 years while average age of Manish and Mansi is 24 years, find the difference between the ages (years) of Mohan and Mansi.

- a. Quantity I = Quantity II or relation cannot be determined
- b. Quantity I > Quantity II
- c. Quantity I $\leq$ Quantity II
- d. Quantity I $\geq$ Quantity II
- e. Quantity I < Quantity II

17. Questions

Mixture A contains milk and water in the ratio of 4: 3 while ratio of milk to water in mixture B is 3: 2. If mixture A and mixture B are mixed in the ratio 7: 10 to form mixture C.

Quantity I: Find the total quantity of mixture C, if difference between quantities of milk and water in mixture C is 5N litres ('N' is single digit number that is a product of two prime numbers).

Quantity II: Find the total quantity of mixture C if difference between quantities of milk and water in mixture C is 6N litres ('N' is single digit prime number not equal to 7).

- a. Quantity I $\geq$ Quantity II
- b. Quantity I = Quantity II or relation cannot be determined
- c. Quantity I > Quantity II
- d. Quantity I < Quantity II
- e. Quantity I $\leq$ Quantity II

18. Questions

Quantity I: Amounts in the ratio of 3: 2 are invested in schemes A and B respectively. Scheme A offers 15% SI and scheme B offers 20% annual CI and total interest from both the schemes at the end of 2 years is ₹7120, then find the difference between SI and CI from both the schemes at the end of 2 years.

Quantity II: An item is marked 66.67% above cost price and sold after two successive discounts of 15% and ₹250 respectively. If total discount given on the item is ₹400, then find the profit amount earned on the item.

If Quantity III = Quantity I + Quantity II and Quantity I _ Quantity III _ Quantity II, then find the symbols that can replace the blanks () in the same order.

- a. <, <
- b. =, <

c. $<, >$

d. $=, >$

e. $<, =$

19. Questions

Below given are three series and each series follow a different logic. Find the missing/wrong terms in the series and answer the question.

I: 43200, 1728, 108, 12, Q, 3

II: $108 \div Q$, 108, 480, 3780, 34020, 374220

III: 12, ?, 1861, 2030, 4774, 4999

What is the difference between the wrong term of series II and missing term of series III?

a. 347

b. 237

c. 391

d. 203

e. 311

20. Questions

If a new series is formed that follows the same logic as in series III and first term is equal to Q, then which of the following cannot be a term of the newly formed series?

a. 124

b. 2021

c. 4992

d. 1852

e. 4765

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Explanations:

1. Questions

Answer: B

The speed of boat A in still water = 14.5 km/h

The speed of boat B in still water = 12 km/h

Let the speed of stream be 'x' km/h.

So, the upstream speed of boat A = $(14.5 - x)$ km/h

And the downstream speed of boat B = $(12 + x)$ km/h

Since, boat A took 3 hours to reach point P and boat B took 2 hours to reach point Q.

So,

$$\Rightarrow (14.5 - x) * 3 + (12 + x) * 2 = 66$$

$$\Rightarrow 43.5 - 3x + 24 + 2x = 66$$

$$\Rightarrow 43.5 + 24 - 66 = x$$

$$\Rightarrow x = 1.5$$

Since, the upstream speed of boat A = $(14.5 - 1.5) = 13$ km/h

So, the distance between points P and X = $13 * 3 = 39$ km

2. Questions

Answer: D

From the equation:

$$\Rightarrow 15x^2 + 26x + 169 = 14x^2 + 48x + 73$$

$$\Rightarrow 15x^2 - 14x^2 + 26x - 48x + 169 - 73 = 0$$

$$\Rightarrow x^2 - 22x + 96 = 0$$

$$\Rightarrow x^2 - 16x - 6x + 96 = 0$$

$$\Rightarrow x(x - 16) - 6(x - 16) = 0$$

$$\Rightarrow x = 16, 6$$

Now, the rate of interest in scheme A = $R = 16\%$

And the rate of interest in scheme B = $r = 6\%$

Since, in both the schemes, interest is compounded half yearly.

So, the CI received from scheme A after 1 year:

$$\Rightarrow 15000 * [1 + (16/2)/100]^2 - 15000$$

$$\Rightarrow 15000 * [1 + 8/100]^2 - 15000$$

$$\Rightarrow 17496 - 15000$$

$$\Rightarrow ₹2496$$

And the CI received from scheme B after 1 year:

$$\Rightarrow 10000 * [1 + (6/2)/100]^2 - 10000$$

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$$\Rightarrow 10000 * [1 + 3/100]^2 - 10000$$

$$\Rightarrow 10609 - 10000$$

$$\Rightarrow ₹609$$

$$\text{Required ratio} = 2496: 609 = 832: 203$$

3. Questions

Answer: B

Let the quantity of type B milk in mixture P = 'x' L

So, the quantity of type A milk in mixture P = (x + 20) L

From the question:

$$\Rightarrow [(x + 20) * 60 + x * 40]/(x + 20 + x) = 55$$

$$\Rightarrow 60x + 1200 + 40x = 110x + 1100$$

$$\Rightarrow 10x = 100$$

$$\Rightarrow x = 10$$

Since, the quantity of type A milk in mixture P = 10 + 20 = 30 L

So, the quantity of type A milk in mixture Q = 30/2 = 15 L

Since, the quantity of type B milk in mixture P = 10 L

So, the quantity of type B milk in mixture Q = 2 * 10 = 20 L

From I:

Since, the difference between the quantities of type A milk and type B milk in mixture Q = 20 - 15 = 5 L

So, statement I is not true.

From II:

The cost of mixture Q = [(15 * 60) + (20 * 40)]/(15 + 20) = ₹48.5 per L (approx.)

Since, the cost of mixture Q is not more than ₹50 per L.

So, statement II is not true.

From III:

The total quantity of mixture P = 30 + 10 = 40 L

The total quantity of mixture Q = 15 + 20 = 35 L

Since, the required ratio = 40: 35 = 8: 7

So, statement III is not true.

Hence, none is true.

4. Questions

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Answer: C

Let the CP of each unit of the article = ₹100p

So, the MP of each unit of the article = 125% of 100p = ₹125p

And the SP of each unit of the article = 90% of 125p = ₹112.5p

Since, the total profit received by selling 'X' units, is equal to the cost price of 1 unit of the article.

So,

$$\Rightarrow (112.5p - 100p) * X = 100p$$

$$\Rightarrow 12.5p * X = 100p$$

$$\Rightarrow X = 8$$

To make neither profit nor loss, the SP of each unit of the article should be = ₹100p

The discount on the marked price should be = 125p - 100p = ₹25p

And the discount percentage should be = $(25p/125p) * 100 = 20\% = (X + 12)\%$

5. Questions

Answer: E

The number of units manufactured by machine A in 1 hour = 25

The number of units manufactured by machine B in 1 hour = X

The number of units manufactured by machine C in 1 hour = (X + 5)

Since, the ratio of the time taken by machine A to manufacture a certain number of units to the time taken by machine C to manufacture the same number of units, is 3: 5.

So, the ratio of the efficiency of machine A to that of machine C = 5: 3

So,

$$\Rightarrow 25/(X + 5) = 5/3$$

$$\Rightarrow 15 = X + 5$$

$$\Rightarrow X = 10$$

The number of units manufactured by machine A in 1 hour = 25 @QuizByMickey

The number of units manufactured by machine B in 1 hour = 10

The number of units manufactured by machine C in 1 hour = (10 + 5) = 15

From I:

The ratio of the efficiency of machine A to that of machine B = 25: 10 = 5: 2

So, the question asked in I can be determined.

From II:

Since, the number of units manufactured by machine B in 1 hour = 10

So, the time taken by machine B to manufacture 300 units of the article = $300/10 = 30$ hours

So, the question asked in II can be determined.

From III:

Since, the total number of units manufactured by machines A, B and C together in 1 hour = $25 + 10 + 15 = 50$ units

So, the total number of units manufactured by machines A, B and C together in 2.4 hours = $50 * 2.4 = 120$ units

So, the question asked in III can be determined.

Hence, all can be determined.

6. Questions

Logic in series I:

$$\Rightarrow 677 - 17^2 = 388$$

$$\Rightarrow 388 - 13^2 = 219$$

$$\Rightarrow 219 - 11^2 = 98$$

$$\Rightarrow 98 - 7^2 = 49$$

$$\Rightarrow 49 - 5^2 = 24$$

So, the value of P = 49

Logic in series II:

$$\Rightarrow 980 - 17^2 = 691$$

$$\Rightarrow 691 - 15^2 = 466$$

$$\Rightarrow 466 - 13^2 = 297$$

$$\Rightarrow 297 - 11^2 = 176$$

$$\Rightarrow 176 - 9^2 = 95$$

So, the value of (P * Q) = 980

And the value of Q = $980/49 = 20$

Now, series III:

$$\Rightarrow (2P/7), (Q + 10), (P + Q - 3), (7Q - 10), (5P - 3Q/4)$$

$$\Rightarrow (2 * 49/7), (20 + 10), (49 + 20 - 3), (7 * 20 - 10), (5 * 49 - 3 * 20/4)$$

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=> 14, 30, 66, 130, 230

Answer: A

Since, the logic in series III:

$$\Rightarrow 14 + 4^2 = 30$$

$$\Rightarrow 30 + 6^2 = 66$$

$$\Rightarrow 66 + 8^2 = 130$$

$$\Rightarrow 130 + 10^2 = 230$$

So, the next term of series III = $230 + 12^2 = 374$

7. Questions

Logic in series I:

$$\Rightarrow 677 - 17^2 = 388$$

$$\Rightarrow 388 - 13^2 = 219$$

$$\Rightarrow 219 - 11^2 = 98$$

$$\Rightarrow 98 - 7^2 = 49$$

$$\Rightarrow 49 - 5^2 = 24$$

So, the value of P = 49

Logic in series II:

$$\Rightarrow 980 - 17^2 = 691$$

$$\Rightarrow 691 - 15^2 = 466$$

$$\Rightarrow 466 - 13^2 = 297$$

$$\Rightarrow 297 - 11^2 = 176$$

$$\Rightarrow 176 - 9^2 = 95$$

So, the value of (P * Q) = 980

And the value of Q = $980/49 = 20$

Now, series III:

$$\Rightarrow (2P/7), (Q + 10), (P + Q - 3), (7Q - 10), (5P - 3Q/4)$$

$$\Rightarrow (2 * 49/7), (20 + 10), (49 + 20 - 3), (7 * 20 - 10), (5 * 49 - 3 * 20/4)$$

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=> 14, 30, 66, 130, 230

Answer: E

From X:

$$P = 49 = 7 * 7$$

$$Q = 20 = 2 * 2 * 5$$

Since, there is no common factor of P and Q, so P and Q are co-prime with each other.

So, X is true.

From Y:

$$\text{Difference of P and Q} = 49 - 20 = 29$$

Since, 29 is a prime number.

So, Y is true.

From Z:

$$\text{Value of } (2P + 5Q) = (2 * 49 + 5 * 20) = 198$$

Since, 198 is not a perfect square.

So, Z is not true.

Hence, only Z is not true.

8. Questions

Answer: C

Logic in series I:

$$5 * 2 + 1 = 11$$

$$11 * 2 - 1 = 21$$

$$21 * 2 + 1 = \mathbf{43(44)}$$

$$43 * 2 - 1 = 85$$

$$85 * 2 + 1 = 171$$

$$171 * 2 - 1 = 341$$

So, the wrong term of series I = P = 44

And the number, which will replace P = A = 43

Logic in series II:

$$1260 \div 4 = 315$$

$$315 \div 3.5 = 90$$

$$90 \div 3 = 30$$

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$$30 \div 2.5 = 12(15)$$

$$12 \div 2 = 6$$

So, the wrong term of series II = Q = 15

And the number, which will replace Q = B = 12

Logic in series III:

$$2^3 + 1 = 9$$

$$3^2 - 1 = 8$$

$$4^3 + 1 = 65$$

$$5^2 - 1 = 24$$

$$6^3 + 1 = 217(210)$$

$$7^2 - 1 = 48$$

So, the wrong term of series III = R = 210

And the number, which will replace R = C = 217

Required ratio:

$$\Rightarrow (P + Q + R) : (A + B + C)$$

$$\Rightarrow (44 + 15 + 210) : (43 + 12 + 217)$$

$$\Rightarrow 269 : 272$$

9. Questions

Answer: C

Let the present ages of A and B be '3x' years and '4x' years respectively.

So, C's present age = $3 * (4x - 3x) = '3x'$ years

Since, the ratio of A's age after 9 years to C's age 6 years ago is 9: 4.

So,

$$\Rightarrow (3x + 9) / (3x - 6) = 9/4$$

$$\Rightarrow 12x + 36 = 27x - 54$$

$$\Rightarrow 15x = 90$$

$$\Rightarrow x = 6$$

Now, A's present age $3 * 6 = 18$ years

B's present age = $4 * 6 = 24$ years

And C's present age = $3 * 6 = 18$ years

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Quantity I:

The average of present ages of A, B and C = $(18 + 24 + 18)/3 = 20$ years

Quantity II:

Since, D is older than A and the difference between the present ages of A and D is 3 years.

So, D's present age will be more than A = $18 + 3 = 21$ years.

Hence, Quantity I < Quantity II

10. Questions

Answer: E

Let the speeds of cars A, B and C be 'a' m/s, 'b' m/s and 'c' m/s respectively.

Since, cars A and B are 780 m away from each other and car A crosses car B in 15 seconds.

So,

$$\Rightarrow (a + b) = 780/15$$

$$\Rightarrow a + b = 52 \text{ -----(1)}$$

Since, cars A and C are 384 m away from each other and car A crosses car C in 8 seconds.

So,

$$\Rightarrow (a + c) = 384/8$$

$$\Rightarrow a + c = 48 \text{ -----(2)}$$

By equation (1) – (2):

$$\Rightarrow a + b - a - c = 52 - 48$$

$$\Rightarrow b - c = 4$$

Quantity I:

Since, car B is following car C.

So, the relative speed of cars B and C = $(b - c)$ m/s

Since, the distance between cars B and C = $780 - 384 = 396$

$$\Rightarrow 396/(b - c)$$

$$\Rightarrow 396/4$$

$$\Rightarrow 99 \text{ seconds}$$

Quantity II:

Since, the value of 'a', 'b' and 'c' can't be determined.

So, the speed of car A can't be determined.

Hence, relation can't be established.

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11. Questions

Answer: A

Since, the ratio of the number of e-bikes to the number of e-scooters sold by company A in March = 4: 5

And the difference between the number of e-bikes and the number of e-scooters sold by company A in March = 7

So, the number of e-bikes sold by company A in March = $7 * 4/(5 - 4) = 28$

And the number of e-scooters sold by company A in March = $7 * 5/(5 - 4) = 35$

Since, the number of e-scooters sold by company A in January = $35 - 5 = 30$

So, the number of e-scooters sold by company B in January = $30 * (5/3) = 50$

The number of e-bikes sold by company A in January = $30 + 6 = 36$

And the number of e-bikes sold by company B in January = $50 - 5 = 45$

Now, the required difference = $45 - 36 = 9$

12. Questions

Answer: E

Since, the ratio of the number of e-bikes sold by company A to that sold by company B in February, is 17: 10.

And the difference between the number of e-bikes sold by companies A and B in February = 21

So, the number of e-bikes sold by company B in February = $21 * 10/(17 - 10) = 30$

Since, the number of e-bikes sold by company B in February is equal to that sold by company A in May.

So, the number of e-bikes sold by company A in May = 30

The number of e-scooters sold by company A in May = $30 + 12 = 42$

And the number of e-bikes sold by company B in May = $30 + 8 = 38$

Since, the number of e-scooters sold by company B in May is 10 less than the number of e-bikes sold by company B in May.

So, the number of e-scooters sold by company B in May = $38 - 10 = 28$

And the required ratio = $42: 28 = 3: 2$

13. Questions

Answer: E

Since, the ratio of the number of e-bikes to the number of e-scooters sold by company B in April = $116(2/3): 100 = 7: 6$

And the difference between the number of e-bikes and e-scooters sold by company B in April = 5

So, the number of e-bikes sold by company B in April = $5 * 7/(7 - 6) = 35$

And the number of e-scooters sold by company B in April = $35 - 5 = 30$

Now, the number of e-bikes sold by company A in April = $35 * (100/87.5) = 40$

And the number of e-scooters sold by company A in April = 120% of 30 = 36

So, the required difference = $40 - 36 = 4$

14. Questions

Answer: B

Since, the number of e-bikes sold by company B in January = 15

So, the number of e-scooters sold by company B in January = $15 + 5 = 20$

Since, the number of e-bikes sold by company B in February = 25

So, the number of e-scooters sold by company B in February = $25 + 9 = 34$

Since, the average number of e-scooters sold by company B in January, February and March = 24

So, the number of e-scooters sold by company B in March = $(3 * 24) - (20 + 34) = 18$

Since, the difference between the number of e-bikes and e-scooters sold by company B in March is 3.

So, the number of e-bikes sold by company B in March will either be $18 - 3 = 15$ or $18 + 3 = 21$.

From I:

If the number of e-bikes sold by company B in March = 15 or 21

Then the number of e-bikes sold by company B in January and March cannot be equal.

So, I cannot be true.

From II:

When the number of e-bikes sold by company B in March = 15

So, the total number of e-bikes and e-scooters sold by company B in March = $15 + 18 = 33$

When the number of e-bikes sold by company B in March = 21

So, the total number of e-bikes and e-scooters sold by company B in March = $21 + 18 = 39$

Since, the total number of e-bikes and e-scooters sold by company B in March can be less than 40.

So, II can definitely true.

From III:

When the number of e-bikes sold by company B in March = 15

So, the average number of e-bikes sold by company B in January, February and March = $(15 + 25 + 15)/3 = 18.33$

When the number of e-bikes sold by company B in March = 21

So, the average number of e-bikes sold by company B in January, February and March = $(21 + 25 + 15)/3 = 20.33$

Since, the average number of e-bikes sold by company B in January, February and March can't be 20.

So, III can't be true.

Hence, only II definitely true.

15. Questions

Answer: D

The ratio of the number of e-bikes to the number of e-scooters sold by company A in February = 100: 72.5
= 1000: 725 = 40: 29

And the difference between them is a prime number.

If we took any values more than (40, 29) i.e; (80, 58), (120, 87), (160, 116) and so on, then the difference between the number of e-bikes and e-scooters sold by company A in February will not be a prime number.

So, the only possible number of e-bikes and e-scooters sold by company A in February are 40 and 29 respectively.

And the difference between the number of e-bikes and e-scooters sold by company A in February = 40 – 29 = 11

16. Questions

Answer: E

Quantity I:

Let time taken by Prateek and Saurabh alone to finish the work is P days and S days respectively.

According to the question:

$$1/24 + 1/P + 1/S = 1/6$$

$$1/P + 1/S = 1/6 - 1/24 = (4 - 1)/24$$

$$1/P + 1/S = 3/24 = 1/8$$

Since Saurabh is 20% more efficient than Prateek which means Prateek alone will take more time than Saurabh alone to finish the work, .

Let,

$$P=5/6S$$

$$(5/6)S+S=1/8$$

$$S=6/88=3/44$$

$$\text{Saurabh finishes the work} = 1/S = 44/3 = 14.66$$

Quantity II:

$$\text{Sum of ages of Manish and Mohan} = 32 * 2 = 64 \text{ years}$$

Let ages of Manish and Mohan is 'x' years and '64 – x' years respectively.

$$\text{Sum of ages of Manish and Mansi} = 24 * 2 = 48 \text{ years}$$

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Age of Mansi = $(48 - x)$ years

Difference between the ages of Mohan and Mansi = $(64 - x) - (48 - x) = 64 - x - 48 + x$

= 16 years

So, Quantity I < Quantity II

17. Questions

Answer: A

Part of milk in mixture A = $4/7$

Part of milk in mixture B = $3/5$

Let part of milk in mixture C = C

By the rule of alligation:

$4/7$ $3/5$

C

7 10

$(3/5 - C): (C - 4/7) = 7: 10$

$6 - 10C = 7C - 4$

$10 = 17C$

$C = 10/17$

Ratio of milk to water in mixture C = 10: 7

Let quantity of milk and water in mixture C is $10x$ and $7x$ respectively.

Quantity I:

Difference between quantities of milk and water in mixture C = $5N$ litres ('N' is single digit number that is a product of two prime numbers).

Single digits prime numbers = 2, 3, 5, and 7.

Product of only 2 and 3 will become single digit number. So, $N = 2 * 3 = 6$

Difference between quantities of milk and water in mixture C = $3x = 5N = 30$ litres

$x = 10$

Total quantity of mixture C = $10x + 7x = 17x = 170$ L

Quantity II:

Difference between quantities of milk and water in mixture C = 6N litres ('N' is single digit prime number not equal to 7.).

Single digits prime numbers ($\neq 7$) = 2, 3, and 5.

When $N = 2$

Difference between quantities of milk and water in mixture C = $3x = 6N = 12$ L

$x = 4$

Total quantity of mixture C = $10x + 7x = 17x = 68$ L

When $N = 3$

Difference between quantities of milk and water in mixture C = $3x = 6N = 18$ L

$x = 6$

Total quantity of mixture C = $10x + 7x = 17x = 102$ L

When $N = 5$

Difference between quantities of milk and water in mixture C = $3x = 6N = 30$ L

$x = 10$

Total quantity of mixture C = $10x + 7x = 17x = 170$ L

So, Quantity I $\geq$ Quantity II

18. Questions

Answer: D

Quantity I:

Let amounts invested in schemes A and B are $3x$ and $2x$ respectively.

According to the question:

$$(3x * 15 * 2)/100 + 2x[(1 + 20/100)^2 - 1] = 7120$$

$$0.9x + 0.88x = 7120$$

$$1.78x = 7120$$

$$x = 4000$$

Difference between SI and CI from both the schemes at the end of 2 years = $(3x * 15 * 2)/100 - 2x[(1 + 20/100)^2 - 1]$

$$= 0.9x - 0.88x$$

$$= 0.02x$$

$$= ₹80$$

Quantity I = 80

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Quantity II:

Let CP of item = $300x$

MP of item = 166.67% of $300x = 500x$

Total discount amount = 15% of $500x + 250 = 400$

$$75x = 150$$

$$x = 2$$

SP of item after two successive discounts = 85% of $500x - 250 = 850 - 250 = ₹600$

CP of item = $300x = ₹600$

Profit amount = $600 - 600 = ₹0$

Quantity II = 0

Quantity III = Quantity I + Quantity II = $80 + 0 = 80$

So, Quantity I = Quantity III > Quantity II

19. Questions

The logic in series I is:

$$43200 \div 5^2 = 1728$$

$$1728 \div 4^2 = 108$$

$$108 \div 3^2 = 12$$

$$12 \div 2^2 = 3 = Q$$

$$3 \div 1^2 = 3$$

So, **Q = 3**

The logic is series II is:

$$108 \div 3, 108, 480, 3780, 34020, 374220$$

$$36 \times 3 = 108$$

$$108 \times 5 = 540 \text{ (Not 480)}$$

$$540 \times 7 = 3780$$

$$3780 \times 9 = 34020$$

$$34020 \times 11 = 374220$$

So, **Wrong term = 480**

The logic in series III is:

$$12 + 11^2 = 133$$

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$$133 + 12^3 = 1861$$

$$1861 + 13^2 = 2030$$

$$2030 + 14^3 = 4774$$

$$4774 + 15^2 = 4999$$

So, **missing term = 133**

Answer: A

Wrong term of series II = 480

Missing term of series III = 133

Required difference = $480 - 133 = 347$

20. Questions

The logic in series I is:

$$43200 \div 5^2 = 1728$$

$$1728 \div 4^2 = 108$$

$$108 \div 3^2 = 12$$

$$12 \div 2^2 = 3 = Q$$

$$3 \div 1^2 = 3$$

So, **Q = 3**

The logic is series II is:

$108 \div 3, 108, 480, 3780, 34020, 374220$

$$36 \times 3 = 108$$

$$108 \times 5 = 540 \text{ (Not 480)}$$

$$540 \times 7 = 3780$$

$$3780 \times 9 = 34020$$

$$34020 \times 11 = 374220$$

So, **Wrong term = 480**

The logic in series III is:

$$12 + 11^2 = 133$$

$$133 + 12^3 = 1861$$

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$$1861 + 13^2 = 2030$$

$$2030 + 14^3 = 4774$$

$$4774 + 15^2 = 4999$$

So, **missing term = 133**

Answer: C

First term of the newly formed series = $Q = 3$

The newly formed series is:

$$3 + 11^2 = 124$$

$$124 + 12^3 = 1852$$

$$1852 + 13^2 = 2021$$

$$2021 + 14^3 = 4765$$

$$4765 + 15^2 = 4990$$

Hence, 4992 cannot be a term of the newly formed series

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